

Detlefs' Harmonic-Quotient Conjecture on OEIS A061002 is Equivalent to Wolstenholme's Theorem

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Abstract

In February 2013, G. Detlefs conjectured on OEIS A061002 that for a prime $p > 3$,

$$\frac{\text{den}(H_p/H_{p-1})}{\text{num}(H_{p-1}/p^2)} = p^3,$$

where $H_n = \sum_{k=1}^n 1/k$. We prove it, and prove the sharper statement that the identity holds for a prime p *if and only if* p^2 divides the numerator of H_{p-1} — that is, the conjecture is not merely a consequence of Wolstenholme's theorem but a restatement of it. Writing $H_{p-1} = A/B$ in lowest terms, the two quantities in the quotient are exactly pA and A/p^2 , and the p^2 in the second is the whole content of the statement. The restriction $p > 3$ is therefore not cosmetic: for $p = 2$ and $p = 3$ the quotient equals 2 and 9, not 8 and 27, because Wolstenholme fails there.

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1 The conjecture

Sequence A061002 of the OEIS [2] is defined by: as p runs through the primes ≥ 5 , a gives $(\text{num } H_{p-1})/p^2$, which the entry notes is an integer by the theorem of Wolstenholme. On 20 February 2013 Gary Detlefs contributed:

Conjecture: If p is the n -th prime and $H(n)$ is the n -th harmonic number, then $\text{denominator}(H(p)/H(p-1))/\text{numerator}(H(p-1)/p^2) = p^3$. $A193758(p)/a(n) = p^3$, $p > 3$.

Here $A193758(n)$ is the denominator of H_n/H_{n-1} . The comment stands unrebutted on the live entry; the page carried the timestamp 22 August 2026. Throughout, num and den mean numerator and denominator of a rational in lowest terms, with positive denominator.

Theorem 1. *Let p be a prime and write $H_{p-1} = A/B$ in lowest terms. Then*

$$\text{den}\left(\frac{H_p}{H_{p-1}}\right) = pA \quad \text{and} \quad \text{num}\left(\frac{H_{p-1}}{p^2}\right) = \frac{A}{\gcd(A, p^2)}.$$

Consequently the quotient of the two equals p^3 if and only if $p^2 \mid A$; and this holds for every prime $p \geq 5$ by Wolstenholme's theorem, but for neither $p = 2$ nor $p = 3$.

2 Two elementary facts

Lemma 2. *For every prime p , $p \nmid B$, where $B = \text{den}(H_{p-1})$.*

Proof. H_{p-1} is a sum of fractions $1/k$ with $1 \leq k \leq p-1$, so it can be written with denominator $L = \text{lcm}(1, \dots, p-1)$. Hence $B \mid L$, and no integer in $\{1, \dots, p-1\}$ is divisible by p , so $p \nmid L$. $\square$

Lemma 3 (Wolstenholme, 1862). *If $p \geq 5$ is prime then p^2 divides $\text{num}(H_{p-1})$. For $p = 2$ and $p = 3$ this fails: $H_1 = 1/1$ and $H_2 = 3/2$, so $A = 1$ and $A = 3$ respectively, and neither is divisible by p^2 .*

3 Proof of Theorem 1

Proof. Write $H_{p-1} = A/B$ with $\gcd(A, B) = 1$, $B > 0$.

The denominator. Since $H_p = H_{p-1} + 1/p$,

$$\frac{H_p}{H_{p-1}} = 1 + \frac{1}{p H_{p-1}} = 1 + \frac{B}{pA} = \frac{pA + B}{pA}.$$

This fraction is already in lowest terms. Indeed $\gcd(pA + B, A) = \gcd(B, A) = 1$, and $\gcd(pA + B, p) = \gcd(B, p) = 1$ by Lemma 2; since every prime divisor of pA divides p or A , we get $\gcd(pA + B, pA) = 1$. Hence $\text{den}(H_p/H_{p-1}) = pA$.

The numerator. $H_{p-1}/p^2 = A/(Bp^2)$. Now $\gcd(A, B) = 1$ and, by Lemma 2, $p \nmid B$; therefore $\gcd(A, Bp^2) = \gcd(A, p^2)$ and $\text{num}(H_{p-1}/p^2) = A/\gcd(A, p^2)$.

The quotient. Dividing,

$$\frac{\text{den}(H_p/H_{p-1})}{\text{num}(H_{p-1}/p^2)} = \frac{pA}{A/\gcd(A, p^2)} = p \cdot \gcd(A, p^2),$$

which equals p^3 exactly when $\gcd(A, p^2) = p^2$, i.e. when $p^2 \mid A$. By Lemma 3 this holds for all primes $p \geq 5$ and fails for $p = 2, 3$, where $\gcd(A, p^2)$ is 1 and 3 and the quotient is therefore 2 and 9. $\square$

Corollary 4 (Detlefs' conjecture). *For every prime $p \geq 5$, $\text{den}(H_p/H_{p-1})/\text{num}(H_{p-1}/p^2) = p^3$; equivalently $A193758(p)/A061002(n) = p^3$ where $p = \text{prime}(n)$.*

Proof. The first claim is Theorem 1 with Lemma 3. For the second, $A193758(p) = \text{den}(H_p/H_{p-1}) = pA$ by the proof above, and $A061002(n) = \text{num}(H_{p-1})/p^2 = A/p^2$ by the definition of the sequence; the quotient is p^3 . $\square$

Remark 5 (the hypothesis is doing all the work). Theorem 1 shows the quotient is $p \cdot \gcd(A, p^2)$ unconditionally. So the conjecture is not a corollary of Wolstenholme's theorem sitting on top of some further identity: it *is* Wolstenholme's theorem, re-encoded as a quotient of a numerator by a denominator. The restriction $p > 3$ that Detlefs attaches is exactly the hypothesis $p \geq 5$ of Wolstenholme, and it cannot be dropped.

Remark 6. Nothing above uses that p^3 rather than p^2 divides $\text{num}(H_{p-1})$ for a Wolstenholme prime such as 16843: since $\gcd(A, p^2) = p^2$ whenever $p^2 \mid A$, the quotient is p^3 for those primes too. The identity does not detect Wolstenholme primes.

4 Computational verification

A short script (Python 3, `fractions` and `sympy`) checks the following; all pass.

1. For every prime $5 \leq p < 300$, computing H_{p-1} and H_p exactly as rationals, the quotient $\text{den}(H_p/H_{p-1})/\text{num}(H_{p-1}/p^2)$ equals p^3 , with no exceptions.
2. For $p = 2$ and $p = 3$ the quotient equals 2 and 9 respectively, not 8 and 27.
3. For every prime $5 \leq p < 300$, $\text{den}(H_p/H_{p-1}) = pA$ and $\text{num}(H_{p-1}/p^2) = A/p^2$, where $A = \text{num}(H_{p-1})$; and $p \nmid \text{den}(H_{p-1})$.
4. For every prime $5 \leq p < 300$, p^2 divides $\text{num}(H_{p-1})$ (Wolstenholme), and the resulting quotients reproduce the first 18 published terms of A061002.

Every numeral in this note is an output of that script.

5 Concluding remarks

The proof is elementary given Wolstenholme's theorem, and we claim no difficulty for it. It is recorded because the comment has stood on A061002 since February 2013 — on an entry whose own definition already cites Wolstenholme — and because the sharper form given here identifies the conjecture as equivalent to that theorem rather than a consequence of it, which is the more useful thing to have on the entry. The natural destination is a comment there rather than a journal.

We have found no earlier proof on the entry or in the literature, but an argument this short could well have been given before, and we would welcome notice of any earlier appearance.

References

- [1] R. Meštrović, *Wolstenholme's theorem: its generalizations and extensions in the last hundred and fifty years (1862–2011)*, arXiv:1111.3057 [math.NT], 2011.
- [2] N. J. A. Sloane et al., *The On-Line Encyclopedia of Integer Sequences*, <https://oeis.org>, 2026. Sequences A061002, A193758, A001008; comment of G. Detlefs, 20 February 2013.